

Sift — Master Specification & Build Status

What this document is. A faithful merge of the two source specifications — **Creator Content Intelligence — v1 spec (rev 6)** and **Sift — The Agent Layer (proposal, rev 2)** — combined into one reference, with everything that has since been added in the codebase, and a complete implementation-status matrix at the end (Part IV). Nothing from either source spec has been dropped; the two are reproduced in full (Parts I and II), Part III records what the build added beyond them, and Part IV maps every feature to its current build state.

Sources: creatorcontentintelligencev6.pdf , siftagentlayerproposalv2.pdf (June 2026).

Part I — Creator Content Intelligence · v1 Specification (rev 6)

A search-first archive for each creator — powered by retrieval, amplified by comment-to-DM, and monetized through demand intelligence. Standalone external platform, field-checked against the 2026 Meta and YouTube API reality.

00 · The thesis

The defensible product is not "AI search for Instagram posts." It is the **feedback loop between audience demand and creator production**. Search gets users in, comment-to-DM amplifies reach, and the searches and comments themselves become the asset: insight into what an audience wants next. Built as **one loop**, not three pillars. The search bar and the answer/chatbot are interfaces; **Demand Radar is the monetizable asset**.

Positioning. Not "an AI chatbot trained on your content." Closer to **Search Console for your creator business** — see what your audience wants, what your content already answers, and what to make next. Ties to ROI: better content, fewer repeated DMs, higher product/affiliate conversion, clearer content calendar.

01 · The product is one loop

A single cycle: (1) audience question → (2) grounded, cited answer → (3) deep link + visit → (4) logged demand signal → (5) creator insight (Demand Radar) → (6) new content → (7) better corpus → (8) better answers. Each turn makes the next stronger.

Feature	Role in the loop	v1 form
Search bar	Interface — the front door (pull)	Plain-language search over the whole archive, with cited results
AI answer / chatbot	Interface — not the star	A grounded answer card (cited, bounded, with a "no strong answer" state) — not open-ended chat
Demand Radar	The monetizable asset	Audience demand turned into evidence cards: what they ask, what is covered, what to make next

Where this sits. Delphi (AI clones; replays the archive, no demand intelligence) · ManyChat & co. (keyword → static link; ours reads the question, retrieves, answers, feeds the radar) · Linktree/Beacons (a menu, not a search engine) · Instagram/Meta AI (optimizes Meta's engagement, exposes no demand data). **The wedge is the loop itself:** none connect what fans ask to what the creator makes next.

02 · Who it's for: the minimum viable creator

≥ **100 evergreen posts** with recurring answerable questions · **visible comment demand** ("where's the post about X?") · ≥ one **monetizable path** (course, coaching, product, affiliate, newsletter) · enough **trust** that followers ask for recommendations · **connectable** (an account you own, or a creator willing to OAuth in and hold a role for v1). Best early niches: high-repetition-question creators — fitness, nutrition, travel, beauty/skincare, creator-business, language teachers, productivity, B2B consultants. Avoid regulated / high-liability categories at first.

03 · Distribution: the link leads, comments amplify

Primary channel — the direct link (front door): bio link (IG allows up to 5) · a permanent **"Search" Highlight** (Story link sticker saved to a Highlight stays clickable indefinitely) · a **search-first landing page** (opens on the search bar) · a **pinned reel** · built for the **in-app browser** (instant load, focused search box, no login wall). The Highlight and pinned reel are audience-onboarding surfaces.

Capture layer — comment-to-DM: for passive/new viewers; a specific CTA ("Comment PLAN and I'll send the exact routine") beats "ask me anything"; when retrieval confidence is low, the DM **falls back to a friendly archive link** — never a shaky answer in the creator's voice. An amplifier, not the front door. **Deep links** tie the two together: the DM link opens the platform with the answer loaded, and those are the same shareable URLs.

04 · Demand Radar, made concrete

Decision-ready: every insight is an **evidence card** turnable into a post idea in under a minute — *Topic · Signal (e.g. 17 searches, 9 comments, +42% WoW) · Current coverage · Audience language · Recommended content · Linked products · Confidence*. **Cold start:** first cards come from the **historical comment backlog**. **Delivery:** push a **weekly digest** (email or DM) — the radar comes to the creator.

05 · Retrieval is the hard part, not the chatbot

Lives or dies on: transcription & OCR coverage · chunking + metadata extraction · **citation accuracy** (a fluent answer that cites the wrong reel is brand risk) · ranking & reranking · a reliable **"no good answer found"** behaviour · freshness & contradiction handling. **Two consequences:** (1) make **citation correctness a first-class metric and a gate** — no comment-to-DM automation goes live until it passes on a held-out set; (2) keep the **retrieval answer separate from the creator-style answer** — correctness, citations, relevance first; voice second.

06 · Scope: ruthless for v1

In scope: the demand loop end-to-end · one owned account, every record creator-scoped · link-first distribution + comment-to-DM capture · comments ingested from day one · an evaluation harness (40–60 labelled questions) from week one · deletion / reindexing + an approval mode for DM · a light monetization

hook (product mentions → creator-approved affiliate links). **Out of scope (defer):** open-ended chat · voice cloning / audio replies · full multi-tenant billing & self-serve onboarding · per-creator custom domains · broad analytics dashboards & mobile apps · a dedicated vector DB (pgvector is plenty).

07 · Architecture and data flow

1. **Onboard** — connect the owned Business/Creator account via OAuth; create namespace + link.
2. **Ingest** — media, captions, metadata, comments from day one; download media; transcribe (Whisper); OCR.
3. **Enrich (LLM)** — summary, topics, keywords, product/location mentions; creator-scoped.
4. **Index** — chunk + embed into pgvector beside relational rows; design for deletion & reindex.
5. **Evaluate** — 40–60 labelled questions run on every OCR/chunking/embedding/reranking change; citation correctness gates DM automation.
6. **Serve** — search-first landing + grounded answer card on one hybrid retrieval layer; every result deep-linkable.
7. **Listen** — a Messaging API webhook receives comments; intent detection routes through the same retrieval layer.
8. **Respond** — public reply + a single private DM (answer + deep link); approval mode first; low-confidence → plain archive link.
9. **Learn** — every query/comment/click clusters into Demand Radar evidence cards.
10. **Monetize** — product mentions map to creator-approved affiliate links, surfaced on results and in answers.

The data reality. IG API with **Instagram Login** (Business/Creator, no Facebook Page; scopes `instagram_business_basic/_manage_comments/_manage_messages/_content_publish`) · **Standard Access** covers owned/app-role accounts (POC starts today); **App Review** gates external creators — submit in parallel, request only scopes the demo uses · **private replies bounded:** one message per comment within 7 days, follow-ups only if the person replies, ~200 DMs/hour/account cap · **comment reads paginated** (~50/page, cursor-based, no timestamp filter — walk cursors, keep a watermark) · **do not scrape** · **YouTube** easy only for owned/authorized channels (caption download needs edit permission).

08 · Recommended stack

Ingestion (IG API via Instagram Login + YouTube Data API) · Transcription/OCR (faster-whisper + PaddleOCR behind one provider interface) · Enrichment + answer card (any frontier LLM behind a thin provider interface, LiteLLM-style) · Embeddings (one versioned model at a time; model+version+dims per chunk) · Storage + vectors (PostgreSQL + pgvector) · Search (**hybrid: Postgres full-text + pgvector cosine, reranked**) · Event analytics (PostHog or plain Postgres event tables) · Evaluation harness · Comment/DM loop (IG Messaging API + webhook) · Per-creator routing (path-based namespaces `yourapp.com/<creator>`) · Backend/jobs (**Python FastAPI + RQ/Celery**; Rust for hot paths) · Frontend (**Qwik / Qwik City**) · Hosting (containers, managed/self-hosted Postgres, S3-compatible storage).

Portability principles (non-negotiable): model-agnostic (every AI call behind a provider interface) · embedding swaps are planned re-embed events · infra-agnostic (Docker, plain Postgres+pgvector, S3) · no silent lock-in.

09 · Build sequence (≈ ten weeks)

Phase	Focus	Exit / gate
0 · Unblock data	Create app; connect owned account (Standard Access); submit App Review in parallel; YouTube key	Media + comments pullable same week
1 · Ingest + evaluate (wk 1–2)	Ingest IG (+ owned YouTube), transcribe, OCR, enrich, embed, store citations; stand up labelled set	100+ items indexed; eval harness running
2 · Search-first link (wk 3–4)	One search box, cited result cards, deep links	Beats native IG search on the labelled set
3 · Demand Radar (wk 5–6)	Cluster comments + searches into evidence cards	Creator marks which of 10 they'd make
4 · Answer card (wk 7)	Grounded, cited card with "no strong answer"	Citation correctness passes the gate
5 · Comment-to-DM (wk 8–9)	Few posts, specific triggers, value-first DM, approval mode	A real comment yields a compliant auto-DM
6 · Monetization proof (wk 10)	Map product mentions to creator-approved links; track clicks	≥ 1 live affiliate link; click data

10 · Metrics that matter

Ingestion (% posts with usable caption/transcript/OCR) · Search quality (top-3 retrieval accuracy on labelled questions) · Answer quality (citation correctness, unsupported-claim rate) · Audience UX (search-to-click, repeat searches, time-to-answer) · Deep links (share/open rate per URL) · Comment-to-DM (trigger rate, DM send success, deep-link click) · Demand Radar (% insights marked "useful") · Creator ROI (posts from insights, affiliate/product clicks, support time saved) · Safety ("no answer found" accuracy, hallucination rate, stale-citation rate). **The two early metrics that decide it:** can a real follower find the right old post faster than through Instagram? Does the creator act on Demand Radar without being persuaded?

11 · The moat: creator demand memory

Embeddings are not a moat. The moat is **creator demand memory** — knowing what a specific creator's audience keeps asking, what content satisfies it, which gaps convert into successful posts, and which products map to those needs. Compounds with audience language, content-gap history, post-performance outcomes, affiliate mappings, and shared deep links.

12 · Risks and how to de-risk

API/App Review dependency (run POC on Standard Access; submit review day one; ingest owned YouTube in parallel) · Weak creator pull (make it part of the weekly content workflow; require Highlight + pinned reel) · Insufficient audience usage (ingest comments from day one) · Hallucination/brand damage (strict grounding, visible citations, citation gate before automation) · Overbuilding the chatbot (cited answer card on proven retrieval, not open chat) · DM-automation compliance (official API only; one value-first reply per comment within 7 days; never unsolicited) · Meta ships it natively (own the cross-platform corpus, creator-side demand intelligence, the affiliate/deep-link layer — the relationship is the account, not the API) · Privacy/GDPR (keep audience queries pseudonymous & aggregate; ship a privacy policy + deletion path before public launch).

13 · Monetization: surfaces and pricing

Every search and DM'd question is **declared intent** — monetize the intent, not the tool. Affiliate mapping is the v1 proof of ROI; the rest are an expansion ladder on the existing retrieval layer.

Surface	What it does	Tag
Email capture bridge	"Get the full checklist by email" on answer cards and in DMs	v1.5
Storefront search	Premium content marked; teaser + route to checkout	v1.5
Embeddable widget	The search box on the creator's site + a "Powered by" footer	v1.5
Insight → draft generator	One tap on a Radar card → reel script, hook, caption	v2
Sponsor pitch reports	Auto one-pager from Radar data ("412 questions about creatine")	v2
Paid Q&A on "no answer"	The failure state becomes revenue; the question still feeds Radar	v2

Pricing tiers (hypothesis): Free (public search page + "Powered by") · Creator (~\$49–99/mo: Demand Radar weekly digest, grounded answer card, affiliate mapping, email capture) · Pro (~\$149–299/mo: comment-to-DM automation, storefront search, draft generator, sponsor reports, multi-platform ingestion). Manual onboarding first.

14 · Build / don't-build decision

Build if: a real follower finds old content faster than native IG search · the creator marks Radar insights genuinely useful and acts · comment-to-DM produces incremental visits without compliance problems. **Pause/pivot if:** search usage stays low after promotion · insights feel obvious · the system cannot reliably cite the right source.

The company is the loop between what an audience asks and what a creator makes next. Search gets them in, comment-to-DM amplifies, and demand intelligence is what creators pay for.

Part II — Sift, the Agent Layer (proposal, rev 2)

The full feature blueprint — from a search tool to a chief of staff for creators. Additive to the v1 spec; v1 ships unchanged; every feature phased v1 → v3.

00 · The shift

v1 shows a creator what their audience wants; an agent acts on it — recommends, drafts, replies, and closes the loop, always with approval. A creator's day comes down to three recurring questions, which the agent answers continuously: **What does my audience want?** (§03) · **What should I make next?** (§04) · **What should I reply to right now?** (§05). Three more domains run the business around the loop: Monetization (§06), Brand & Reputation (§07), Growth & Strategy (§08), Operations & Ecosystem (§09). The missing pieces over v1 are **agency** and **proactivity**.

01 · The agent loop

Listen (capture every search & comment — v1) → **Sift** (cluster intent, strip noise, spot trends) → **Serve** (grounded cited answer to the fan — v1) → **Advise** (turn demand into a drafted recommendation) → **Execute** (draft content, replies, links for one-tap approval) → **Close** (tell the original asker their question was answered, and measure whether it worked). The loop closes when the asker hears back, seeding the next question.

02 · Foundations the agent learns from

Feature	What it does	Tag
Demand clustering & dedup	Merges semantically similar questions into one cluster ("142 asked this"). Underpins every radar view	v1
Voice memory	Stores approved replies/hooks and draft edits so generated content stays on-voice	v1 capture
Offer-awareness	Creator tags products/offers/campaigns; Sift routes every CTA/affiliate suggestion to the right one	v1.5
Rules & preferences	Tone, taboo topics, escalation rules, monetization priorities	v1.5
Demand-item state machine	Each "useful" card moves Idea → Drafting → Published → Loop-closed	v1
Intent / outcome event log	Logs who asked what, what was served, what the creator did, what happened next	v1

The only thing that touches v1: build the state machine and the intent/outcome event log into the v1 schema now, plus start capturing demand clusters and voice examples. Cheap, and the difference between a feature set and a rebuild.

03 · Audience & demand intelligence

Feature	What it does	Tag
The Briefing	A pushed digest: top signals with ready hooks in voice, a re-promotion pick, open gaps, trend/resurfacing alerts, urgent inbox replies, one fully drafted post	v1.5
Content gap map	What the creator has answered well / partially / contradicted / never addressed — a strategic map	v1.5
Persona segmentation	Clusters the audience by behaviour, gives each segment its own demand feed, flags one-group bias	v2
Sentiment & emotion	Reads emotional tenor (anxiety, confusion, excitement, frustration)	v2
Trend detection	Rising themes and old posts suddenly searched; weekly, with spike alerts above a threshold	v2
Cross-platform demand synthesis	Ingests newsletter replies, forwarded DMs, podcast comments; answers "asked everywhere?" and "asked only on YouTube?"	v2
Competitor demand radar	Opt-in, aggregate: how a similar creator answered the same demand	v3

04 · Content production

Feature	What it does	Tag
Content briefs	One-page brief from a demand cluster: exact phrasing, coverage gaps, hook + CTA, source-post links	v1.5
Draft generator (script + hooks)	Ready-to-shoot reel script + hooks grounded in the creator's content and audience's literal words	v1.5
Creator recall search	Creator-only "where did I say that?" — exact sentence, link, reuse ideas	v1.5
Multi-format repurposing + Sift & Shift	One piece → IG carousel, X thread, newsletter, TikTok hooks; "Sift & Shift" migrates a YouTube-only topic to an IG carousel	v2
Series builder	Repeated demand → a planned arc: parts, FAQ post, DM follow-up, offer tie-in	v2
Performance prediction	Scores a draft against the creator's own top-10% structural patterns	v3
Content calendar / planner	Cadence-aware queue: format per topic, posting window, over-saturation flags, reactive window	v3
Thumbnail & visual intelligence	Thumbnail concepts + scoring against historical CTR	v3

05 · Engagement & inbox

Feature	What it does	Tag
Intent-labelled queue	One prioritised queue: content request, purchase intent, support, collab lead	v1.5
Bounded clarifying question	One either/or to disambiguate a vague search — not open chat	v1.5
No-answer waitlist	Low-confidence → "want a heads-up when Alex covers this?" Captures an email, feeds the gap map	v1.5
Saved playbooks	"When I get this kind of question, do this" templates	v1.5
Reply assistant	Drafts public/DM replies + follow-ups, grounded in citations, in approved tone	v2
Agentic DM (approval mode)	Multi-turn DM within Meta's one-message/7-day limit; bulk-approve daily; "always approve this type"	v2
Community delegation	When a follower answered well, like their comment instead of spending the single-DM quota	v2
"Not now, but..." queue	Defer a valid question; Sift holds it, drafts when due, re-contacts the asker within limits	v2
Loop-Closer	When the creator publishes what an asker requested, notify the original asker (compliant channels only)	v2

06 · Monetization & revenue

Feature	What it does	Tag
Auto-affiliate injection	At caption time, append the mapped affiliate link + UTM	v1.5
Offer-aware CTAs	Attach the right current offer/affiliate CTA to answers and recommendations	v1.5

Feature	What it does	Tag
Affiliate optimisation	A/B-test placement, anchor text, pairings; surface winners	v2
Sponsor matchmaker & pitch	Demand → proof for brands; match interests to programs; draft a pitch in the audience's language	v2
Dynamic pricing & offer insights	Pre-purchase questions, objections, bundle signals, price-sensitivity by segment	v3
Revenue forecasting & goals	Project income across affiliate/products/sponsorships; flag off-track; turn a goal into a plan	v3
Brand partnership portal	A two-sided surface where brands discover creators by audience profile	v3 · scale

07 · Brand & reputation

Feature	What it does	Tag
Crisis / sentiment-shift detection	Watch for negative-sentiment spikes, controversy, external mentions; alert early with a response framework	v2
Voice consistency scoring	Score new content (esp. from contractors) against the voice model, with deviation notes	v2
Plagiarism / unauthorised-use monitoring	Detect reposts/scrapers; offer takedown + outreach	v3

08 · Growth & strategy

Feature	What it does	Tag
Content strategy advisor	Strategic questions grounded in Sift's data: expand vs double down, segment growth, evergreen vs trend balance	v2
Peer benchmarking	Contextualise against peers (opt-in/aggregate): what top performers cover that they don't	v3
Education & skill planning	Skill gaps from performance patterns; a 1/3/5-year development plan	v3

09 · Operations & ecosystem

Feature	What it does	Tag
Task / project integration	Radar card → tracked task synced with Notion/Asana/Linear; completion links to the published post	v3
Customer CRM	Unified customer profiles for paid products; lifecycle marketing	v3
Team & role-based access	Owner / content-creator / analytics-only (VA) / contractor roles + approval workflows + audit trail	v3
Creator marketplace	Collaboration discovery by audience overlap; creator-to-creator affiliate	v3 · scale

An API for third-party integrations on Sift's intelligence layer

Feature	What it does	Tag
Public API & dev ecosystem		v3 · scale

10 · What to avoid (and why)

Feature	Why excluded	Tag
Open-ended chat	Hallucination/brand risk; the bounded answer card captures the value safely	avoid
Auto-posting without approval	Too much trust too early; approval mode stays default through v3	avoid
Real-time trend alerts	Low signal-to-noise for one creator; revisit above a query-volume threshold	threshold
Native video editing	Scope creep; production stays at scripts/hooks/clip references	avoid
Cross-platform identity resolution	Privacy-radioactive under GDPR; do topic-level synthesis, never individual matching	privacy

Gated on scale, not the calendar: the brand partnership portal, creator marketplace, and public API only matter at multi-creator scale — pursue once the single-creator loop is proven and repeatable.

11 · Guardrails that carry over

Retrieval correctness is **the gate** (citation correctness on a held-out set before any automation) · **voice is a post-retrieval rewrite** (rewrite a correct cited answer; never generate novel claims) · **approval mode by default** · **DM limits shape the Close step** (one creator-initiated message per comment within 7 days) · **privacy stays aggregate and opt-in** (pseudonymous & clustered demand; opt-in cross-platform/competitor; deletion path + policy before launch).

12 · Positioning

From "Search Console for your creator business" (passive, analytical) to **"Your AI chief of staff — Sift listens to your audience, tells you what they want, and helps you build it."** Short: *"The memory and the roadmap for your creator business."*

13 · Sequencing

Phase	What lands
v1 · unchanged	Search-first archive, Demand Radar, comment-to-DM. Groundwork: state machine, intent/outcome log, demand clustering, voice-memory capture
v1.5 · assist	The Briefing, content gap map, content briefs, draft + hook generator, creator recall, bounded clarifying, no-answer waitlist, intent-labelled inbox, saved playbooks, auto-affiliate, offer-aware CTAs, offer/rules tagging
v2 · agent	Reply assistant, agentic DM, community delegation, Loop-Closer, "not now" queue, persona segmentation, sentiment, trend detection, cross-platform synthesis, repurposing + Sift & Shift, series builder, sponsor matchmaker + pitch, affiliate optimisation, strategy advisor, crisis detection, voice scoring
v3 · scale	Performance prediction, content calendar, thumbnail intelligence, dynamic pricing, revenue forecasting, customer CRM, task/PM integration, team/roles, peer benchmarking, education/skill planning, competitor radar, plagiarism monitoring, brand portal, marketplace, public API

Part III — What the build added (beyond the two specs)

The implementation kept the entire spec above and went further on three axes the source docs only gestured at. These are **built and tested**, not planned.

A · Cross-platform connector framework

The specs assume Instagram (+ owned YouTube). The build generalised this into a **connector contract + capability registry** so product logic asks what an account can *do* rather than assuming a uniform loop:

- **Capabilities** — `content.read`, `media.read`, `captions.read`, `comments.read`, `comments.reply`, `messages.send`, `demand.read`, `analytics.read`, `events.webhook`, `content.publish`, `deletions.receive`.
- **Native connectors** (injectable HTTP transport → tested against fakes, no network): **YouTube** (Data API v3 — uploads-playlist walk, comment threads, replies, both paginated, 403-resilient), **Instagram** (Graph API — media + comments backfill, capped cursor walk, replies), **TikTok** (Display API v2 — *archive* mode = content only, vs *full-loop* = comments + replies where approved; **per-account** capability via `PlatformAccount.full_loop`).
- **Demand sources** — newsletter / podcast / Discord as `demand.read`-only inputs that feed Demand Radar (this is the spec's §03 *cross-platform demand synthesis*, realised via an `ExternalSignal` → `radar` path).
- **sync_native worker** — maps any connector's output into Posts + Comments idempotently (PII-redacted, pseudonymised, question/sentiment-classified, per-item SAVEPOINT isolation), stamps `last_synced_at`. Driven **manually** ("Sync now"), **all-at-once** ("Sync all"), or **hourly** by the scheduler.
- **Canonical answers** (`CanonicalContent`) — variant grouping across platforms; the basis for **Sift & Shift** and per-platform demand segmentation (`everywhere` / `platform:x` / `search_only`).

B · Demand-integrity & explainability (hardening the "asset")

- **Demand Integrity** — unique askers, organic vs CTA-prompted, persistence, sentiment, intent, **manipulation risk** (how inflatable), **exposure-normalized demand** (asks per 1k impressions).
- **Explainable Opportunity Score** — components exposed, not a black box.
- **Reconciliation** — every radar card classified against the creator's own archive: `well` / `partial` / `gap`, with the covering permalink surfaced ("you've already answered this — re-promote" vs "genuine gap — make it").
- **North-star metric** — closed loops / creator / week, broken down **made vs re-promoted**; **re-promote** is a second loop-closure path.
- **Versioned claim layer** (freshness/contradiction), **causal layer** (interventions, holdouts, baselines, confidence intervals, holdout lift), **trust firewall** (why-selected + disclosure flags), **Demand Certificate**, canonical knowledge endpoint.

C · Productionisation & security

- SQLAlchemy 2.0 models, all creator-scoped; **Alembic** migrations 0001–0004; `EncryptedString` (Fernet/MultiFernet) for OAuth tokens; opt-in RLS.
- **Billing** — Plan enum (free/creator/pro), feature gating, quotas; Stripe behind an interface + offline fake. **AI cost metering** across agent generators with op tags + per-plan cost cap; a **separate anonymous-answer abuse ceiling** so the free answer card stays free but a fan page can't drive unbounded LLM cost.
- GDPR export/erase + PII redaction · **dedicated pseudonym secret** (decoupled from the admin token) · webhook idempotency + signature **fail-closed** in production · **fail-closed boot** on insecure default secrets · **prompt-injection hardening** (untrusted audience text wrapped + guarded on every agent LLM call) · rate limiting · RQ job reliability · **DM-dispatch advisory lock** (no double-send / cap-bypass).
- CI/CD + observability; k8s + Render deploy manifests.
- **Offline by default** — Fake providers for LLM/embeddings/transcription/OCR/rerank/ billing, so the whole system runs and tests with no network or API keys.

D · Surfaces

- **Fan pages** (Qwik, near-zero-JS) — search-first, grounded answer card, deep links.
- **Creator Studio** (Qwik SSR) — opportunity queue ranked by Opportunity Score, **filterable by platform × gap/answered**, with per-card actions (make / draft / series / re-promote / ignore); the **Platforms** page (connectors + capabilities, connection + last-synced status, Sync now / Sync all, canonical answers, Sift & Shift, per-platform top demand); Inbox, Drafts (deep-linked), Briefing, north-star breakdown.

E · Test & quality posture

311 Python tests (against real Postgres + pgvector) + **9 frontend tests** (Vitest), ruff-clean, four migrations applying cleanly, and a five-dimension adversarial code review (security/tenancy, persistence/migrations, API surface, workers/connectors, retrieval/agent) with all findings fixed.

Part IV — Implementation status (every feature)

Legend: **built** · **partial / basic** · **not built (deferred)** · **deliberately avoided**.

v1 spec (rev 6)

§	Item	Status	Where / note
01	Search bar (hybrid, cited)		<code>cci_retrieval</code> (FTS + pgvector + RRF + rerank)
01	Grounded answer card ("no strong answer")		<code>answer.py</code>
01	Demand Radar (evidence cards)		<code>radar.py</code> , <code>demand.py</code>
03	Search-first landing + deep links		<code>apps/web</code> fan pages, <code>deep_links.py</code>
03	Comment-to-DM capture + low-confidence fallback		<code>dm.py</code> (approval mode)

§	Item	Status	Where / note
04	Cold start from comment backlog		<code>build_radar(backlog=True)</code>
04	Weekly digest delivery		<code>digest.py</code> + scheduler
05	Transcription / OCR / chunking / rerank		providers + <code>cci_retrieval</code>
05	Citation correctness gate		<code>cci_eval</code> harness (now fails closed on vacuous pass)
06	Eval harness (40–60 labelled)		<code>cci_eval</code>
06	Deletion / reindex + approval mode		<code>indexer.py</code> , GDPR, DM approval
06	Affiliate links (light monetization)		products + affiliate redirect
07	Onboard → ... → Monetize (10-step flow)		end-to-end
07	Instagram Login / Standard Access / scopes		client + OAuth flow built; no live consent round-trip (env-bound)
08	Hybrid search, Postgres+pgvector, Qwik, FastAPI+RQ		as specified
08	Provider interfaces / portability principles		<code>cci_providers</code> (all swappable, offline fakes)
10	Metrics + events		<code>events.py</code> , metrics endpoints
12	Privacy/GDPR (pseudonymous, deletion, policy)		<code>privacy.py</code> , <code>gdpr.py</code> , legal docs
13	Email capture bridge		<code>WaitlistEntry</code>
13	Storefront search / embeddable widget		deferred (premarket)
13	Pricing tiers (free/creator/pro)		<code>billing.py</code> Plan enum + gating

Agent layer (rev 2)

§	Feature	Tag	Status	Where / note
02	Demand clustering & dedup	v1		<code>radar.py</code>
02	Voice memory	v1		<code>VoiceExample</code> , <code>capture_voice</code>
02	Offer-awareness	v1.5		<code>Offer</code> , offer-aware CTAs
02	Rules & preferences	v1.5		<code>CreatorRules</code>
02	Demand-item state machine	v1		<code>DemandState</code> + transitions
02	Permission ladder	—		<code>permissions.py</code> — per-action levels, GET/PUT / <code>permissions</code> , pause switch
02	Intent / outcome event log	v1		<code>Outcome</code> , <code>Event</code>
03	The Briefing	v1.5		<code>briefing.py</code>
03	Content gap map	v1.5		<code>content_gap_map</code> + coverage/reconciliation
03	Persona segmentation	v2		<code>intelligence.py</code>
03	Sentiment & emotion	v2		<code>score_sentiment</code>

§	Feature	Tag	Status	Where / note
03	Trend detection	v2	▪	<code>intelligence.py</code> (weekly; threshold-gated spikes)
03	Cross-platform demand synthesis	v2	▪	demand sources + <code>ExternalSignal</code> → radar
03	Competitor demand radar	v3	▶ ▪	<code>growth.competitor_radar</code> — opt-in gate + endpoint live; aggregates are placeholders pending a multi-creator panel
04	Content briefs	v1.5	▪	<code>generate_brief</code>
04	Draft generator (script + hooks)	v1.5	▪	<code>generate_draft</code> (cost-metered)
04	Creator recall search	v1.5		<code>creator_recall</code>
04	Multi-format repurposing + Sift & Shift	v2		<code>repurposing.py</code>
04	Series builder	v2		<code>build_series</code> (persists draft stubs)
04	Performance prediction	v3	▶ ▪	<code>predict_performance</code> (basic; needs history to be reliable)
04	Content calendar / planner	v3	▶ ▪	<code>content_calendar</code> (basic)
04	Thumbnail & visual intelligence	v3	▶ ▪	<code>thumbnail_concepts</code> (concepts; no CTR model)
05	Intent-labelled queue	v1.5		<code>label_message</code> , <code>/inbox</code>
05	Bounded clarifying question	v1.5	▪	<code>bounded_clarify</code> — in the public search response on no-strong-answer; fan page renders the either/or chips
05	No-answer waitlist	v1.5		<code>WaitlistEntry</code> + fan-page email form on no-strong-answer (shows cohort size)
05	Saved playbooks	v1.5		<code>Playbook</code> CRUD + <code>match_playbook</code> applied in the comment → DM pipeline (matched template, still approval mode)
05	Reply assistant	v2	▶ ▪	<code>engagement.draft_reply</code> — drives the agentic-DM path (itself ▶); comment-DM uses the grounded answer card directly
05	Agentic DM (approval mode)	v2	▶ ▪	<code>handle_dm_followup</code> , <code>can_auto_approve</code> , <code>bulk_approve</code> (thread dispatcher not wired)
05	Community delegation	v2	▪	<code>delegation_candidate</code> — surfaced as a hint in the DM approval queue (the like itself is manual: no official comment-like API)
05	"Not now, but..." queue	v2	▪	<code>defer_question</code> , <code>DeferredItem</code> , <code>due_deferrals</code>
05	Loop-Closer	v2	▪	<code>close_loop</code> + re-promote/outcomes
06	Auto-affiliate injection	v1.5	▪	<code>inject_affiliate</code> applied at draft approval (caption-time proxy — auto-posting is deliberately avoided)
06	Offer-aware CTAs	v1.5		offers → CTAs
06	Affiliate optimisation	v2		<code>affiliate_optimisation</code>
06	Sponsor matchmaker & pitch	v2		<code>sponsor_report</code>
06	Dynamic pricing & offer insights	v3	▶	<code>pricing_insights</code> (basic)

§	Feature	Tag	Status	Where / note
06	Revenue forecasting & goals	v3	▸	<code>revenue_forecast</code> (basic)
06	Brand partnership portal	v3·scale		deferred (scale-gated)
07	Crisis / sentiment-shift detection	v2	▸	<code>brand.py</code> (sentiment-shift; no external-web watch)
07	Voice consistency scoring	v2		<code>brand.py</code>
07	Plagiarism / unauthorised-use monitoring	v3	▸	<code>growth.plagiarism_scan</code> — real pgvector-similarity repost detector via endpoint; no external-web watch
08	Content strategy advisor	v2		<code>strategy_advisor</code>
08	Peer benchmarking	v3	▸	<code>growth.peer_benchmark</code> — opt-in gate + endpoint live; peer norms are placeholders pending a real panel
08	Education & skill planning	v3	▸	<code>education_plan</code> (basic)
09	Task / project integration	v3	▸	<code>growth.export_task</code> endpoint (task payload export); no external PM sync
09	Customer CRM	v3	▸	<code>/customers</code> endpoint (basic profiles)
09	Team & role-based access	v3		<code>TeamMember</code> + roles + <code>ApiKey</code> + audit log
09	Creator marketplace	v3·scale		deferred (scale-gated)
09	Public API & dev ecosystem	v3·scale	◦	<code>public_api.py</code> <code>/v1/public/*</code> with scoped API keys
10	Open-ended chat	avoid	◦	grounded answer card instead
10	Auto-posting without approval	avoid	◦	approval mode default
10	Real-time trend alerts	threshold	▸	threshold-gated only
10	Native video editing	avoid		scripts/hooks only
10	Cross-platform identity resolution	privacy	▸	topic-level synthesis only
11	Guardrails (gate, voice-rewrite, approval, DM limits, privacy)	—		enforced throughout (citation gate, post-retrieval rewrite, approval mode, 7-day/cap, pseudonymity)

Build additions (Part III) — all

Connector framework + capability registry · YouTube/Instagram/TikTok native connectors · TikTok per-account full-loop · demand sources → radar · `sync_native` (manual / all / scheduled) + last-synced · canonical answers + per-platform segmentation · reconciliation + re-promote + closed-loop breakdown · demand integrity (manipulation risk, exposure-normalized) · opportunity score · claim layer · causal layer · trust firewall · demand certificate · knowledge endpoint · billing + quotas + cost metering + abuse ceiling · GDPR + pseudonym secret · fail-closed secrets/webhook · injection hardening · DM advisory lock · migrations 0001–0004 · Studio dashboard + fan pages · 311 + 9 tests, five-dimension review.

Summary

Every **v1** and **v1.5** feature is **built**; the **v2** agent layer is **built**; **v3** is **mostly built** (basic implementations of prediction, calendar, thumbnails, pricing, forecasting, education, CRM; team/roles and public API fully built). The scale-gated items (competitor radar, peer benchmarking, plagiarism scan, task export) exist as **opt-in-gated scaffolds with live endpoints** (placeholder aggregates pending a multi-creator panel); only the **brand portal and creator marketplace** remain fully deferred, and the **deliberately-avoided** items remain correctly unbuilt. Beyond the specs, the build added a full cross-platform connector framework, demand-integrity and explainability layers, and a production security/billing/observability posture.